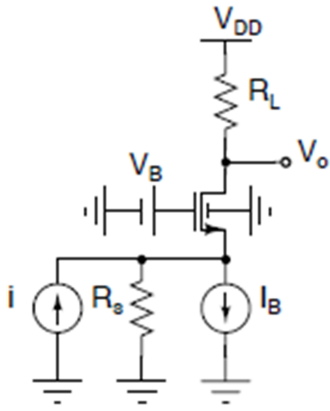


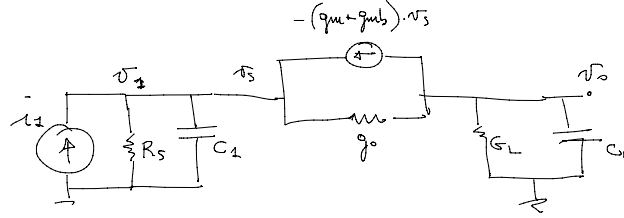
Lecture-22

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S.S.d.

$$C_1 = C_{gs} + C_{sb}$$



$$v_1 = v_3$$

$$g_o = \frac{1}{r_o}$$

$$Y = \begin{pmatrix} y_i & y_r \\ y_p & y_o \end{pmatrix}$$

$$y_i = \frac{i_1}{v_1} \Big|_{v_o=0}$$

$$y_r = \frac{i_1}{v_3} \Big|_{v_o=0}$$

$$y_p = \frac{i_o}{v_1} \Big|_{v_o=0}$$

$$y_o = \frac{i_o}{v_o} \Big|_{v_1=0}$$

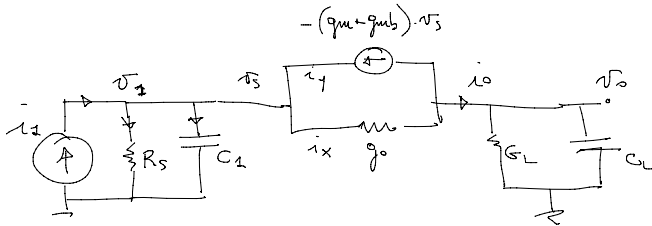
$$Y = \begin{pmatrix} G_s + sC_1 + \cancel{g_o} + g_m + g_{mb} & -\cancel{g_o} \\ -\cancel{g_o} - g_m - g_{mb} & G_L + sC_L + \cancel{g_o} \end{pmatrix}$$

$$G_t = g_o + g_{mb} + g_m$$

voltage
Gain

=

$$A_v = -G_s \cdot \frac{y_p}{\Delta Y} = \frac{G_s \cdot G_t}{(G_s + G_t + sC_1)(G_L + g_o + sC_L) - G_t \cdot g_o}$$



$$\left. \begin{aligned} g_m + g_{mb} &\triangleq g_{m+} \\ Z_{in} &\triangleq \frac{R_s}{1 + sC_1 R_s} \end{aligned} \right\} \text{Assumptions}$$

$$i_1 = \frac{v_1}{Z_{in}} + g_{m+} \cdot v_1$$

$$i_o = g_{m+} \cdot v_1 \quad (i_y \gg i_x)$$

$$\Rightarrow i_o = i_1 \left(\frac{Z_{in}}{1 + g_{m+} Z_{in}} \right) \cdot g_{m+}$$

$$\frac{i_o}{i_1} = \frac{1}{\frac{1}{Z_{in} \cdot g_{m+}} + 1} \approx \frac{1}{1 + s \frac{C_1}{g_{m+}}}$$

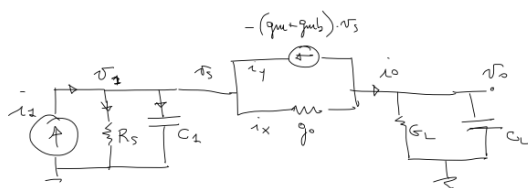
$$\frac{i_o}{i_1} \approx 1 \quad (\text{DC})$$

$$Z_{in} = \frac{R_s}{1 + sC_1 R_s} \approx \frac{1}{sC_{in}}$$

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→ In previous lecture we calculated the "Voltage Gain" of the "Common Gate" stage in terms of Frequency Response.

∴ The main Role of CG stage is as a Current Buffer let's calculate i/p Resistance & Current Gain for the circuit.



$$\left. \begin{aligned} g_m + g_{mb} &\triangleq g_{m+} \\ Z_{in} &\triangleq \frac{R_S}{1 + sC_1 R_S} \end{aligned} \right\} \text{Assumptions}$$

$$\left. \begin{aligned} i_1 &= \frac{v_1}{Z_{in}} + g_{m+} \cdot v_1 \\ i_o &= g_{m+} \cdot v_1 \quad (i_y \gg i_x) \end{aligned} \right\} \Rightarrow i_o = i_i \left(\frac{Z_{in}}{1 + g_{m+} Z_{in}} \right) \cdot g_{m+}$$

$$\frac{i_o}{i_i} = \frac{1}{\frac{1}{Z_{in} \cdot g_{m+}} + 1} \approx \frac{1}{1 + s \frac{C_1}{g_{m+}}}$$

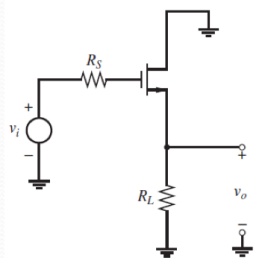
$$\left(\frac{i_o}{i_i} \approx 1 \quad (\text{DC}) \right) \downarrow \text{DC gain}$$

$$\left(Z_{in} = \frac{R_S}{1 + sC_1 R_S} \approx \frac{1}{sC_{in}} \right) \rightarrow \text{approximation}$$



Common Drain Frequency Response

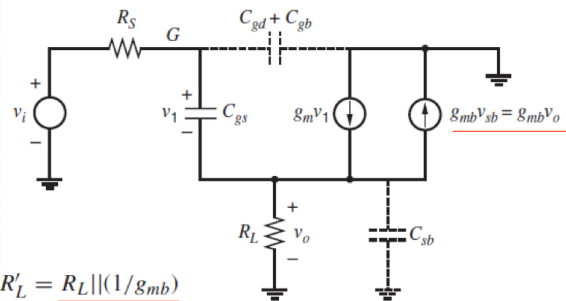
Frequency Response of Source Followers



C_{gd} , C_{gb} , and C_{sb} are ignored

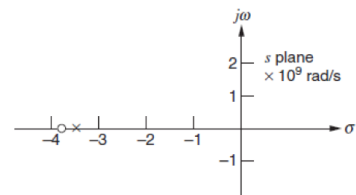
$$z_1 = -\frac{g_m}{C_{gs}} \approx -\omega_T$$

$$p_1 = -\frac{1}{C_{gs}R_1}$$

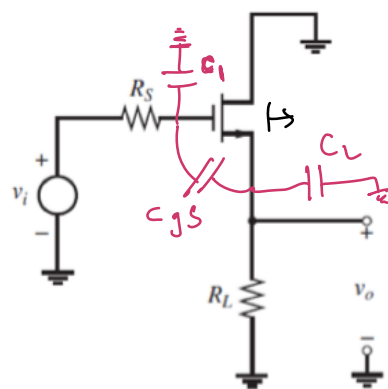


$$R'_L = R_L || (1/g_{mb})$$

$$R_1 = \frac{R_S + R'_L}{1 + g_m R'_L}$$



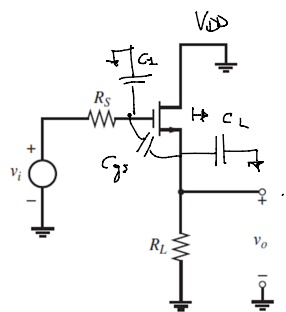
16



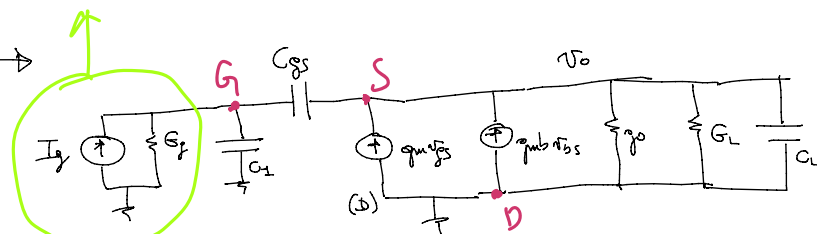
C_1 is because of C_{GD}

Common Drain Frequency Response

Norton's Equivalent of v_i & R_s



S.S.C



$$C_1 = C_{gs} + C_{gd} ; C_L = C_L' + C_{sb}$$

$$\rightarrow g_m v_{gs} = g_m v_g - g_m v_s$$

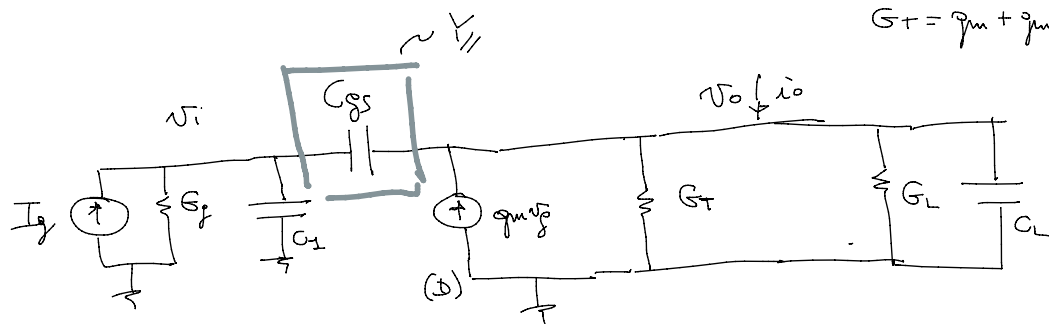
$$v_s = v_o$$

$$g_m v_{gs} = g_m v_g - g_m v_s = g_m v_g - g_m v_o$$

$$g_m v_{gs} = g_m v_g - g_m v_o$$

$$C_1 = C_{gs} + C_{gd}$$

$$C_L = C_L' + C_{sb}$$



$$G_T = g_m + g_{mb} + g_o$$

$$y_i = G_s + sC_1 + sC_{gs}$$

$$y_r = \frac{i_i}{v_o} \Big|_{v_i=0} = 0 - sC_{gs}$$

$$y_f = -g_m - sC_{gs}$$

$$y_o = G_T + G_L + sC_L + sC_{gs}$$

$$Z(s) = \frac{v_o}{I_g} = -\frac{y_f}{\Delta Y} = \frac{g_m + sC_{gs}}{\Delta Y}$$

i/p because we have taken Norton's equivalent of v_i & R_s

$$\Delta Y = [G_s + s(C_1 + C_{gs})][G_T + G_L + sC_L + sC_{gs}] - sC_{gs}(g_m + sC_{gs})$$

$$Z(s) = \frac{g_m}{G_s(G_L + g_m)} \cdot \frac{1 + \frac{sC_{gs}}{g_m}}{1 + as + bs^2}$$

$$1 + as + b \cdot s^2 \triangleq 1 + \frac{s}{\omega_0 Q} + \frac{s^2}{\omega_0^2}$$

$$a \triangleq \frac{1}{\omega_0 Q} \Rightarrow Q = \frac{1}{\omega_0 a}$$

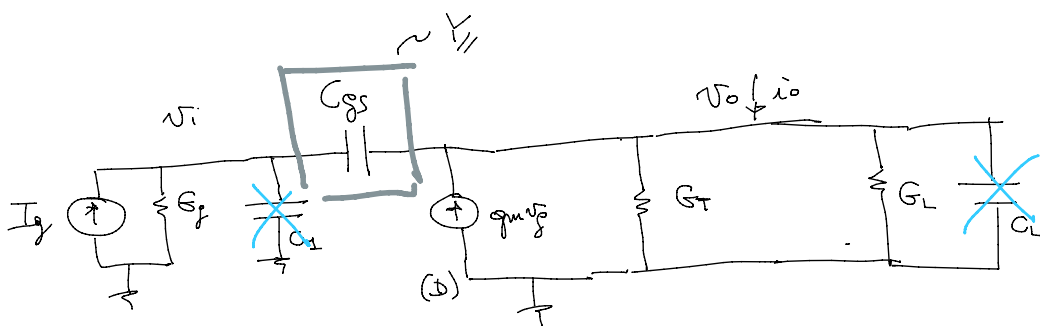
$$b = \frac{1}{\omega_0^2} \Rightarrow \omega_0 = \frac{1}{\sqrt{b}}$$

$$\omega_0 = \frac{1}{\sqrt{b}} = \sqrt{\frac{G_g (g_m + G_L)}{C_1 (G_L + C_{gs}) + C_L \cdot g_g}}$$

$$Q = \frac{\sqrt{(C_1 C_L + C_1 \cdot C_{gs} + G_L \cdot g_g) G_g \cdot (g_m + G_L)}}{G_g (G_L + C_{gs}) + G_L (C_1 + C_{gs}) + g_m \cdot C_1}$$

$$p_{1,2} = \frac{-\frac{1}{\omega_0 Q} \pm \sqrt{\left(\frac{1}{\omega_0^2 Q^2} - \frac{4}{\omega_0^2}\right)}}{\frac{2}{\omega_0^2}}$$

$Q > \frac{1}{2}$ complex conj. poles



$$y_i = G_g + \cancel{sC_1} + sC_{gs}$$

$$y_r = \frac{i_1}{v_o} \Big|_{v_{in} \rightarrow 0} = 0 - sC_{gs}$$

$$y_f = -g_m - sC_{gs}$$

$$y_o = G_T + G_L + \cancel{sC_L} + sC_{gs}$$

$$G_L' = G_L + g_{mb} + g_o$$

$$\begin{aligned} \Delta Y &= (G_g + sC_{gs})(G_T + G_L + sC_{gs}) - (sC_{gs} + g_m) \cdot sC_{gs} \\ &= G_g(G_T + G_L) + sC_{gs}(G_T + G_L + G_g) + \cancel{s^2 C_{gs}^2} - \cancel{s^2 C_{gs}^2} - g_m sC_{gs} \\ &= G_g(g_m + G_L') + sC_{gs}(G_L' + G_g) \end{aligned}$$

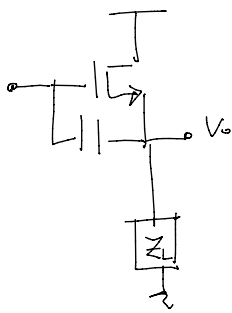
$$\Delta Y = G_g (g_m + G_L') \left[1 + \frac{sC_{gs} (G_L' + G_g)}{G_g (g_m + G_L')} \right]$$

$$-\frac{y_f}{\Delta Y}$$

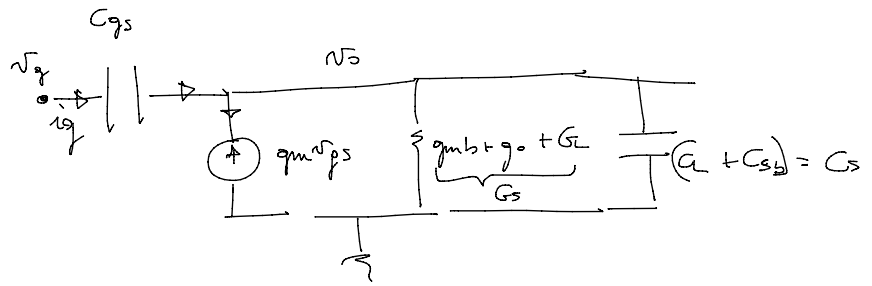
$$sC_{gs} \cdot \frac{G_L' R_g + 1}{g_m + G_L'} = sC_{gs} \frac{R_g + R_L'}{g_m R_L' + 1}$$

$$\phi_1 = - \frac{1}{C_{gs}} \frac{1 + g_m R_L'}{R_g + R_L'}$$

→ pole of the Transfer function for C_0



S.S.C



$v_g ; i_g$

$$i_g = (v_g - v_o) \cdot sC_{gs}$$

$$(v_g - v_o) \cdot sC_{gs} = -g_m(v_g - v_o) + (G_s + sC_S) \cdot v_o$$

$$v_g(sC_{gs} + g_m) = v_o(G_s + sC_S + g_m + sC_{gs})$$

$$v_o = \frac{v_g(sC_{gs} + g_m)}{G_s + g_m + sC_S + sC_{gs}} \quad ; \quad i_p = v_g \left(1 - \frac{sC_{gs} + g_m}{g_m + G_s + s(C_S + C_{gs})} \right) \cdot sC_{gs}$$

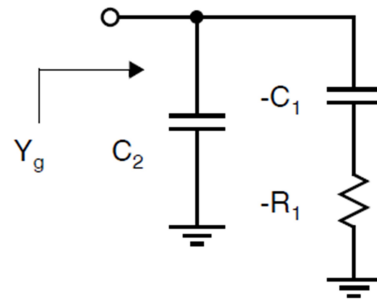
$$\frac{i_g}{v_g} = Y_g = \frac{sC_{gs} \cdot (sC_S + G_s)}{s(C_S + C_{gs}) + g_m + G_s}$$

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Source Follower Input Impedance

$$Y_g = \frac{i_{g1}}{V_{g1}} = \frac{sC_{gs1}(sC_s + G_{s1})}{s(C_{gs1} + C_s) + g_{m1} + G_{s1}}$$

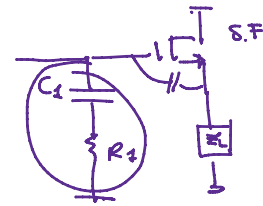
$$= sC_2 + \frac{1}{-R_1 - \frac{1}{sC_1}}$$



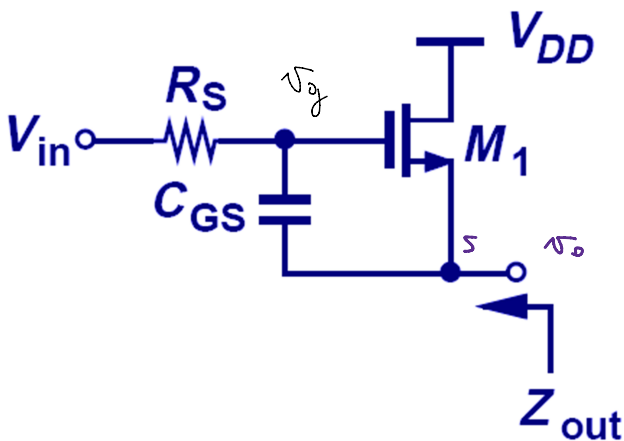
$$\rightarrow C_1 = \frac{C_{gs1}(C_s g_{m1} - C_{gs1} G_{s1})}{(g_{m1} + G_{s1})(C_{gs1} + C_s)} \cong \frac{g_{m1} C_{gs1} C_s}{(g_{m1} + G_{s1})(C_{gs1} + C_s)}$$

$$\rightarrow R_1 = \frac{(C_{gs1} + C_s)^2}{C_{gs1}(C_s g_{m1} - C_{gs1} G_{s1})} \cong \frac{(C_{gs1} + C_s)^2}{C_{gs1} C_s g_{m1}}$$

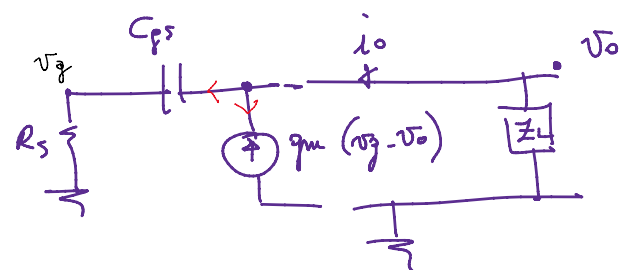
$$\rightarrow C_2 = \frac{C_{gs1} C_s}{C_{gs1} + C_s}$$



Johns - Martin



$$\xrightarrow{\text{S.S.C}} \frac{v_o}{i_o} \bigg|_{v_i=0} = Z_o$$



$$i_o = -g_m (v_g - v_o) + sC_{gs} (v_o - v_g) \quad \left. \vphantom{i_o} \right\}$$

$$(v_o - v_g) \cdot sC_{gs} = \frac{v_g}{R_s}$$

$$v_o = \frac{\frac{v_g}{R_s} + v_g \cdot sC_{gs}}{sC_{gs}} = v_g \left(1 + \frac{1}{R_s \cdot sC_{gs}} \right) = v_g \left(\frac{R_s \cdot sC_{gs} + 1}{R_s \cdot sC_{gs}} \right) = v_o$$

$$(v_g - v_o) = v_g \left(1 - \frac{R_s \cdot sC_{gs} + 1}{R_s \cdot sC_{gs}} \right) = -v_g \left(\frac{1}{R_s \cdot sC_{gs}} \right)$$

$$i_o = \left(\frac{g_m}{R_s \cdot sC_{gs}} + \frac{1}{R_s} \right) \cdot v_g$$

$$i_o = \left(\frac{g_m}{R_s \cdot sC_{gs}} + \frac{1}{R_s} \right) \left(\frac{R_s \cdot C_{gs} \cdot s}{1 + R_s \cdot C_{gs} \cdot s} \right) \cdot v_o$$

$$i_o = \left(\frac{g_m R_s + sC_{gs}}{\cancel{R_s \cdot sC_{gs}}} \right) \left(\frac{\cancel{R_s} \cdot \cancel{C_{gs}} \cdot s}{1 + R_s C_{gs} \cdot s} \right) \cdot v_o$$

$$\frac{i_o}{v_o} = \frac{g_m + sC_{gs}}{1 + R_s \cdot C_{gs} \cdot s} \Rightarrow \frac{v_o}{i_o} = \frac{1 + R_s sC_{gs}}{g_m + sC_{gs}} = Z_o$$

$$Z_o = \frac{1}{g_m} = \frac{(1 + R_s \cdot sC_{gs}) g_m - (g_m + sC_{gs})}{g_m (g_m + sC_{gs})} =$$

$$= \frac{\cancel{g_m} + g_m \cdot R_s \cdot sC_{gs} - (\cancel{g_m} + sC_{gs})}{D(s)} = \frac{\cancel{g_m} (R_s sC_{gs} - \frac{sC_{gs}}{\cancel{g_m}})}{\cancel{g_m} (g_m + sC_{gs})} =$$

$$= \frac{sC_{gs} \left(R_s - \frac{1}{g_m} \right)}{g_m + sC_{gs}} \triangleq Z(s)$$

$$Z_o = Z(s) + \frac{1}{g_m}$$

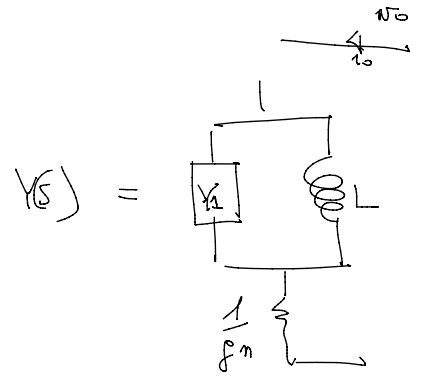
$$Z(s) \triangleq \frac{1}{Y(s)}$$

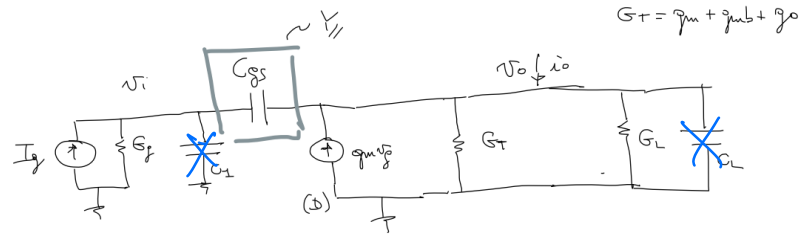
$$Y(s) = \frac{g_m + s C_{gs}}{s C_{gs} \left(R_S - \frac{1}{g_m} \right)} = Y_1 + \frac{1}{sL}$$

$$Y_1 = \frac{1}{R_S - \frac{1}{g_m}}$$

$$L = \frac{C_{gs}}{g_m} \left(R_S - \frac{1}{g_m} \right)$$

$$R_S > \frac{1}{g_m}$$





$$y_i = G_g + sC_{gd} + sC_{gs}$$

$$y_r = \left. \frac{i_i}{v_o} \right|_{v_i=0} = 0 - sC_{gs}$$

$$y_f = -g_m - sC_{gs}$$

$$y_o = G_T + G_L + sC_{gd} + sC_{gs}$$

→ In many cases the most relevant is C_{gs}
we have learned that $C_{gs} > C_{gd}$ in saturation

→ For this reason C_L can be neglected.

$$C_{gs} > C_{gd}$$

$$C_{sb} < C_{gs} \therefore C_L \text{ can be neglected}$$

$$\begin{aligned} \Delta Y &= (G_g + sC_{gs})(G_T + G_L + sC_{gs}) - (sC_{gs} + g_m) \cdot sC_{gs} \\ &= G_g(G_T + G_L) + sC_{gs}(G_T + G_L + G_g) + \cancel{s^2 C_{gs}^2} - \cancel{s^2 C_{gs}^2} - g_m sC_{gs} \\ &= G_g(g_m + G_L') + sC_{gs}(G_L' + G_g) \\ &= G_g(g_m + G_L') \left[1 + \frac{sC_{gs}(G_L' + G_g)}{G_g(g_m + G_L')} \right] \end{aligned}$$

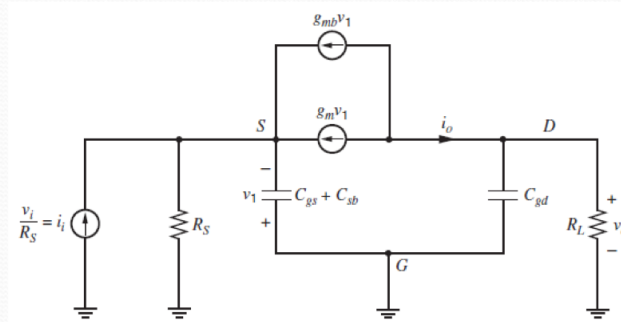
$$G_L' = G_L + g_{mb} + g_o$$

$$\frac{C_{gs} C_{gs} (G_L' + G_g)}{G_g (g_m + G_L')}$$

$$- \frac{y_f}{\Delta Y}$$



Common-Gate Frequency Response



$$\frac{i_o}{i_i} = \frac{1}{1 + s \frac{C_{gs} + C_{sb}}{g_m + g_{mb}}}$$

$$p_1 = -(g_m + g_{mb}) / (C_{gs} + C_{sb})$$

→ pole of common Gate

If $C_{gs} \gg C_{sb}$

$$|p_1| \approx > g_m / C_{gs} \approx \omega_T$$

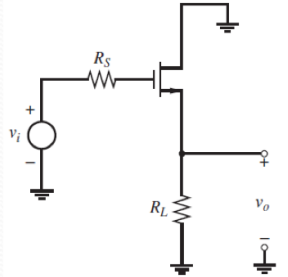
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∴ The transconductance is typically High value
the pole occurs typically at High frequencies
and this means that Common Gate Amplifier
has very good "Frequency Response".

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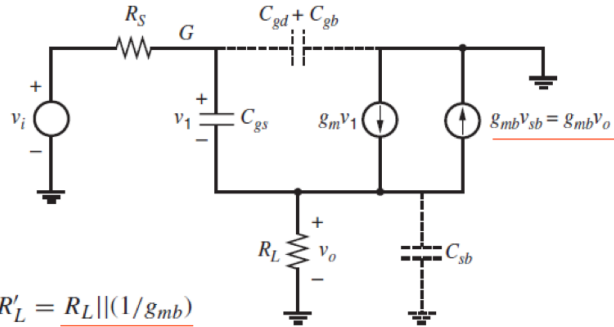
Frequency Response of Source Followers



C_{gd} , C_{gb} , and C_{sb} are ignored

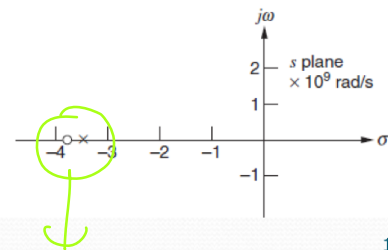
$$z_1 = -\frac{g_m}{C_{gs}} \approx -\omega_T$$

$$p_1 = -\frac{1}{C_{gs}R_1}$$



$$R'_L = R_L || (1/g_{mb})$$

$$R_1 = \frac{R_S + R'_L}{1 + g_m R'_L}$$



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pole and zero are closely located $\therefore$ zero somehow compensates the pole.

Ex:-

Calculate the transfer function for a source follower with $C_{gs} = 7.33$ pF, $k'W/L = 100$ mA/V², $R_L = 2$ k Ω , $R_S = 190$ Ω , and $I_D = 4$ mA. Ignore body effect, and let $C_{gd} = 0$, $C_{gb} = 0$, and $C_{sb} = 0$.

From the data, $g_m = \sqrt{2(100)4}$ mA/V = 28.2 mA/V. Ignoring body effect, we have $R'_L = R_L || (1/g_{mb}) = R_L$. Since $C_{gd} = 0$, ω_T of the device is

$$f_T = 612 \text{ MHz}$$

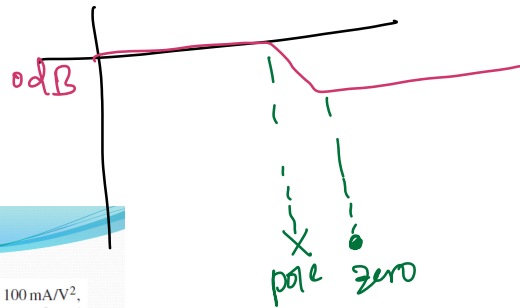
$$\omega_T = \frac{g_m}{C_{gs}} = \frac{28.2 \frac{\text{mA}}{\text{V}}}{7.33 \text{ pF}} = 3.85 \times 10^9 \text{ rad/s}$$

$$z_1 = -\frac{g_m}{C_{gs}} = -3.85 \times 10^9 \text{ rad/s}$$

$$p_1 = -\frac{1}{7.33 \times 10^{-12} \times 38.2} \text{ rad/s} = -3.57 \times 10^9 \text{ rad/s}$$

$$R_1 = \frac{190 + 2000}{1 + 0.0282 \times 2000} \Omega = 38.2 \Omega$$

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→ earlier we have calculated Frequency Response of Common Drain by neglecting C_1 & C_L

* If they were included into calculation.

When capacitors C_{gd} , C_{gb} , and C_{sb} that were ignored above are included

$$\frac{v_o}{v_i} = \frac{g_m R'_L}{1 + g_m R'_L} \frac{1 + s \frac{C_{gs}}{g_m}}{1 + as + bs^2} = A(0) \frac{N(s)}{1 + \frac{s}{\omega_0 Q} + \frac{s^2}{\omega_0^2}}$$

$$a \approx \frac{R'_L(C_{gs} + C_{sb}) + R_S C_{gs} + R_S g_m R'_L C'_{gd}}{1 + g_m R'_L} \quad b \approx \frac{R_S R'_L [C_{sb} C_{gs} + C_{gs} C'_{gd}]}{1 + g_m R'_L}$$

parameter ω_0 is the resonant frequency and parameter Q is the Q-factor. It is well known that if $Q < \sqrt{1/2} \approx 0.707$, then the magnitude of the transfer-function will have its maximum at dc and there will be no peaking (assuming the zero is at a very-high frequency and therefore has negligible effect). Furthermore, for $Q = \sqrt{1/2}$, the -3dB frequency is equal to ω_0 . When the time-domain response is investigated, restrictions on the Q-factor can also be found to guarantee no overshoot for a step input. Specifically, for there to be no overshoot in the step-response, it is necessary that both poles be real which is equivalent to the requirement that $Q \leq 0.5$. In the case where $Q > 0.5$, the percentage overshoot of the output voltage can be shown to be given by

$$\% \text{ overshoot} = 100 e^{-\frac{\pi}{\sqrt{4Q^2 - 1}}}$$

large load and/or input capacitor), then Q becomes small and there will be no overshoot (though the circuit will be slow)

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*** Possibility of using Time constant method to find poles in Common Drain stage.

It is different in the case of Common Drain because it is difficult to apply Miller's Theorem to CD stage.

Because the simplifying assumptions of Miller theorem somehow are nullified or if there is no capacitive effect.

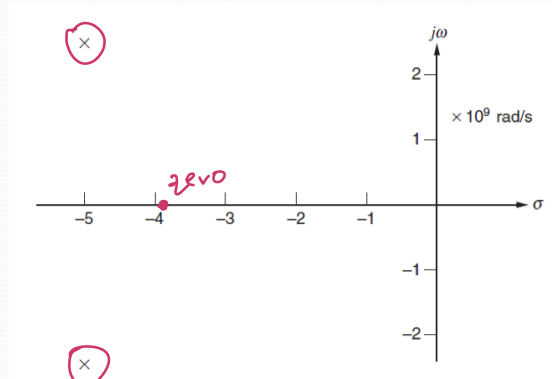
$\therefore$ we can't rely on Miller theorem for CD

Miller's Theorem works pretty well when Gain is very high.

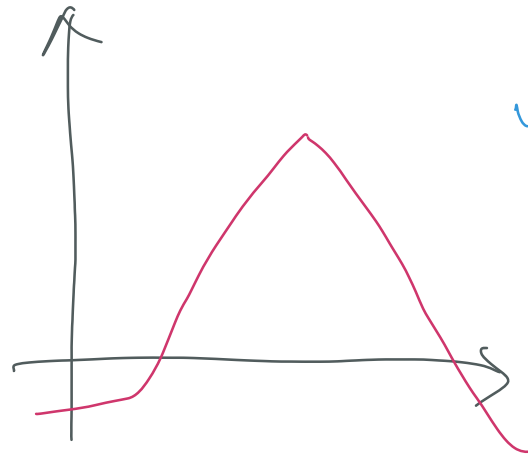
Calculate the transfer function for a source follower with $C_{gs} = 7.33$ pF, $C_{gd} = 0.1$ pF, $C_{gb} = 0.05$ pF, $C_{sb} = 0.5$ pF, $k'W/L = 100$ mA/V², $R_L = 2$ k Ω , $R_S = 190$ Ω , and $I_D = 4$ mA. Ignore body effect.

$$z = -\frac{g_m}{C_{gs}} = -\frac{28.2 \frac{\text{mA}}{\text{V}}}{7.33 \text{ pF}} = -3.85 \times 10^9 \text{ rad/s}$$

$$p_{1,2} = -5.1 \times 10^9 \pm j2.3 \times 10^9 \text{ rad/s}$$



complex conjugate poles



we have Band Pass
kind of Response

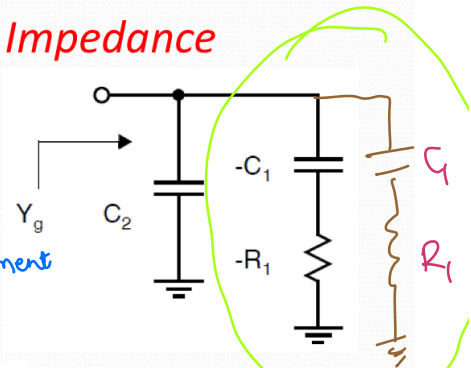


Source Follower Input Impedance

$$Y_g = \frac{i_{g1}}{V_{g1}} = \frac{sC_{gs1}(sC_s + G_{s1})}{s(C_{gs1} + C_s) + g_{m1} + G_{s1}}$$

$$= sC_2 + \frac{1}{-R_1 - \frac{1}{sC_1}}$$

This is just
Rearrangement
of terms



$$C_1 = \frac{C_{gs1}(C_s g_{m1} - C_{gs1} G_{s1})}{(g_{m1} + G_{s1})(C_{gs1} + C_s)} \approx \frac{g_{m1} C_{gs1} C_s}{(g_{m1} + G_{s1})(C_{gs1} + C_s)}$$

$$R_1 = \frac{(C_{gs1} + C_s)^2}{C_{gs1}(C_s g_{m1} - C_{gs1} G_{s1})} \approx \frac{(C_{gs1} + C_s)^2}{C_{gs1} C_s g_{m1}}$$

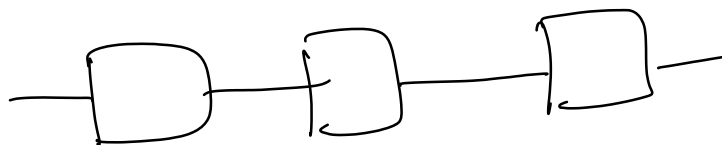
$$C_2 = \frac{C_{gs1} C_s}{C_{gs1} + C_s}$$

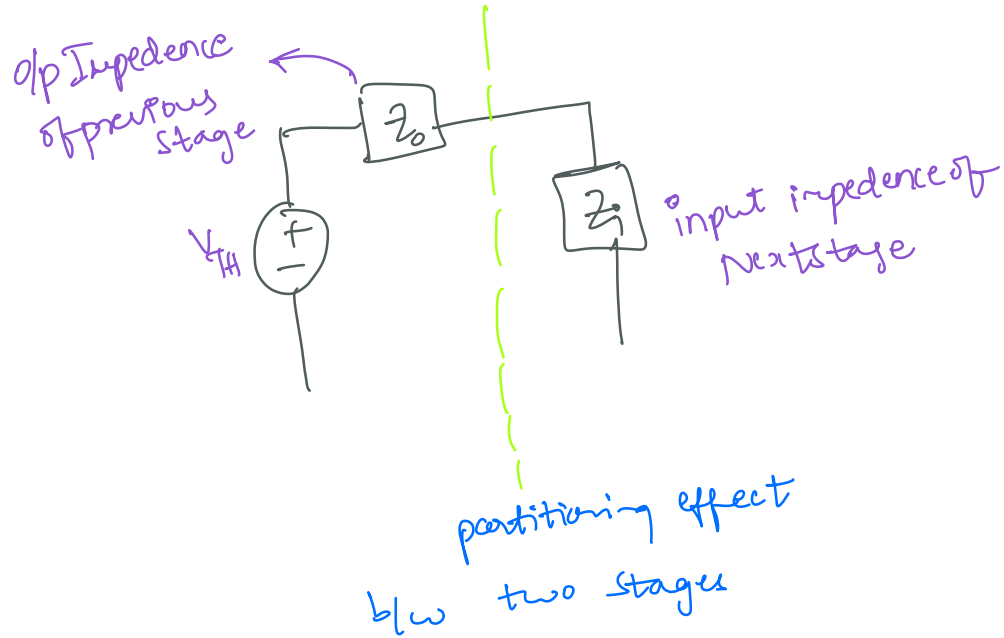
By adding C_1, R_1 in
parallel to the circuit
we can cancel out

The effect of $-C_1$ & $-R_1$
which are imaginary
terms.



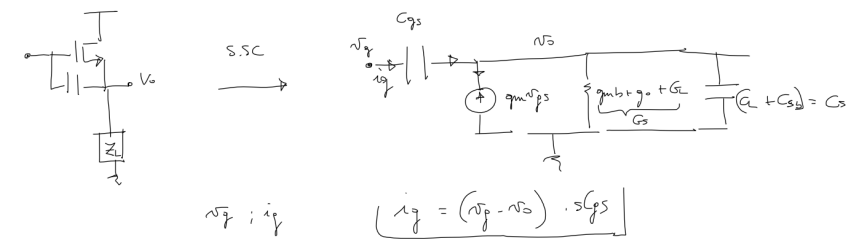
In circuits multiple stages are cascaded





→ we know
The gain of CD is more or less equal to one.

Calculation of Input Impedence of CD stage



$$(v_g - v_o) \cdot s C_{gs} = -g_m (v_g - v_o) + (G_s + s C_s) \cdot v_o$$

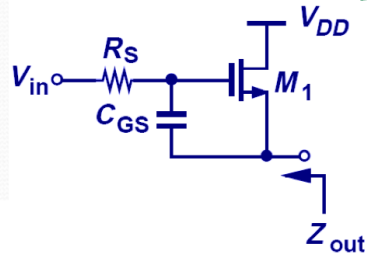
$$v_g (s C_{gs} + g_m) = v_o (G_s + s C_s + g_m + s C_{gs})$$

$$v_o = \frac{v_g (s C_{gs} + g_m)}{G_s + g_m + s C_s + s C_{gs}} \quad \rightarrow \quad i_g = v_g \left(1 - \frac{s C_{gs} + g_m}{g_m + G_s + s(C_s + C_{gs})} \right) \cdot s C_{gs}$$

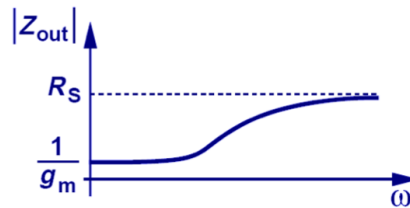
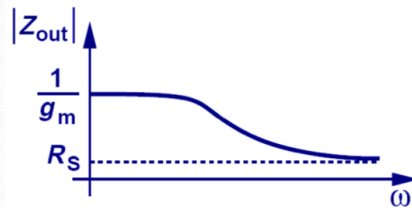
$$\frac{i_g}{v_g} = Y_g = \frac{s C_{gs} \cdot (s C_s + G_s)}{s(C_s + C_{gs}) + g_m + G_s}$$



Source-Follower Output Impedance

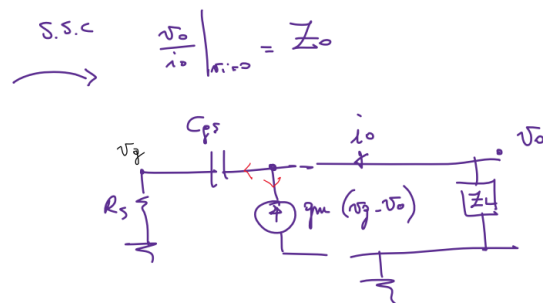
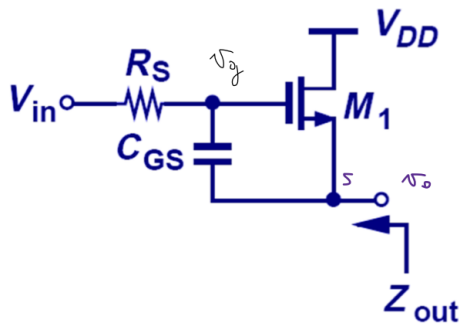


$$\frac{V_X}{I_X} = \frac{R_S C_{GS} s + 1}{C_{GS} s + g_m}$$



Since a follower's primary duty is to lower the driving impedance ($R_S > 1/g_m$), the "active inductor" characteristic on the right is usually observed

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